



EXTREMAL GRAPH THEORY

Erdős Problem 915

From Proof to Search

Ten-minute version

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The question

How many edges can a graph on n vertices carry
if *no* pair of vertices has m edge-disjoint paths?

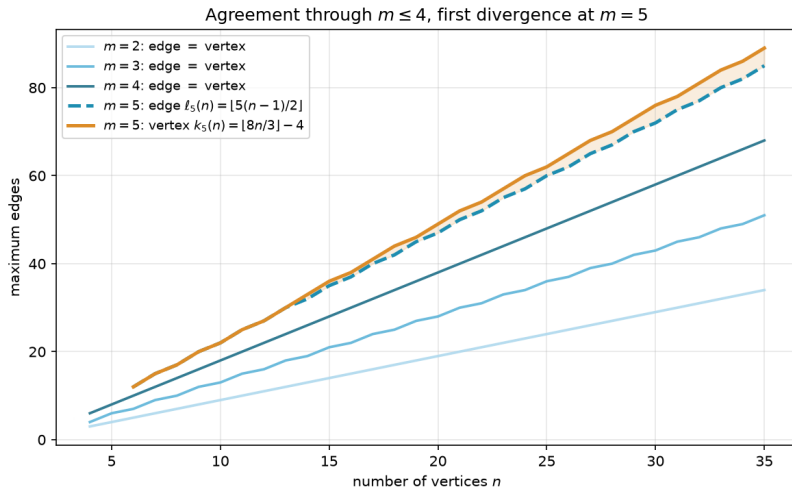
$$\ell_m(n) = \max \{ |E(G)| : |V(G)| = n, \lambda_G^{\max} \leq m - 1 \}$$

Mader settled it for simple graphs, $n \geq m$:

$$\ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$$

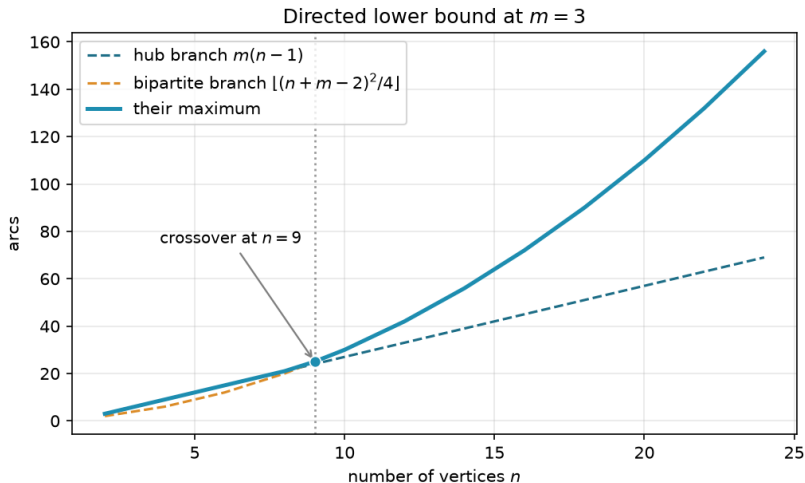
This thesis moves that question along three axes: model, direction, separation. Twelve variants.

Surprise one: edge and vertex routes part company



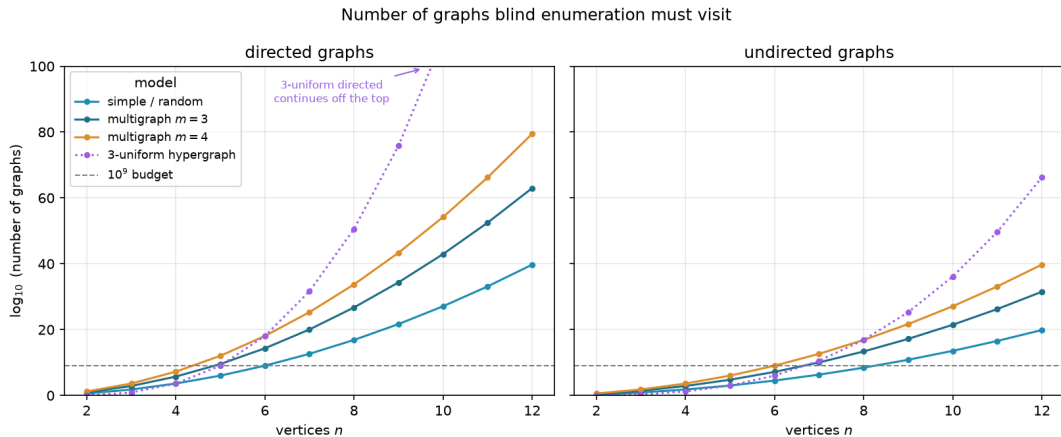
Identical through $m = 4$. At $m = 5$ the vertex value pulls above the edge value, and past $m = 5$ it has been open since 1974.

Surprise two: one-way traffic buys a quadratic



Undirected graphs are stuck at $\Theta(n)$ edges. A digraph reaches $\Theta(n^2)$ arcs at the same connectivity ceiling. The two branches cross at $n = 9$ when $m = 3$.

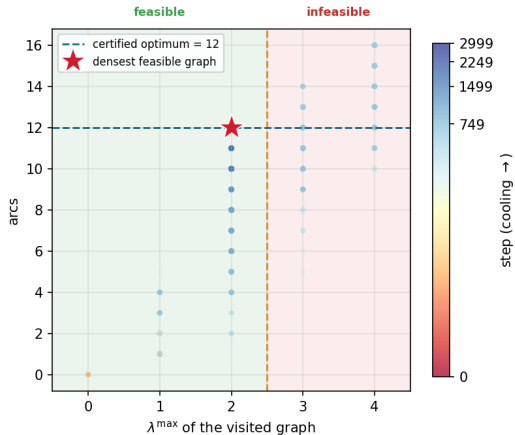
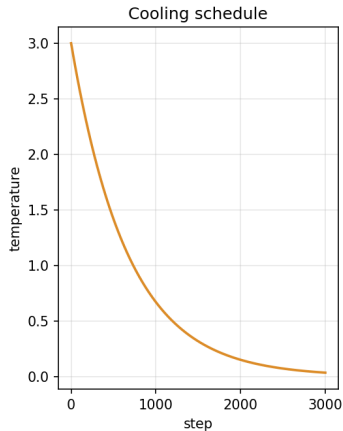
Why a machine: enumeration detonates



Checking every graph passes any reachable budget while n is still in single digits. The directed cases, exactly the open ones, go first.

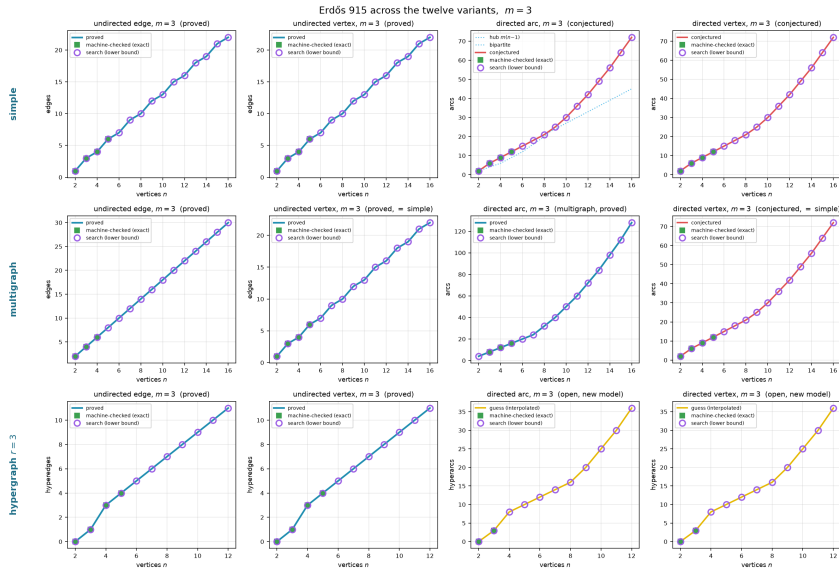
What the machine does: cool a random graph into shape

Cooling toward an extremal graph



Every visited graph, coloured by step. Hot: it strays above the ceiling. Cool: it settles onto the densest feasible graph, here the proved optimum of 12 arcs.

All twelve variants at once



Blue proved for all n , red conjectured, yellow interpolated. Squares proved exactly, circles are lower bounds.

Where each variant stands

Separation	Model	Value	Status
Undirected			
edge	simple	$\left\lfloor \frac{m(n-1)}{2} \right\rfloor$	PROVED Mader
edge	multi	$(m-1)(n-1)$	PROVED
edge	hyper	$\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$	PROVED Gomory–Hu
vertex	simple	$\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ ($m \leq 4$), $\lfloor 8n/3 \rfloor - 4$ ($m = 5$, $n \geq 6$, $n \neq 7, 12$)	CITED open $m \geq 6$
vertex	multi	adjacency count, so the simple value	convention
vertex	hyper	$\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ ($m = 2$, and $m = 3$ in the stated construction range)	BOUND $m = 3$, open $m \geq 4$
Directed			
arc	simple	$\max(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor)$ ($m = 2$), $n^2/4 + \Theta_m(n)$ ($m \geq 3$)	PROVED $m = 2$, CONJ $m \geq 3$
arc	multi	$(m-1) \max(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor)$	PROVED every n and m
arc	hyper	$(1 + o(1))(m-1)n^2/(4(r-1))$	LEADING TERM exact value open
vertex	simple	$\max(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor)$ ($m = 2$), $n^2/4 + \Theta_m(n)$ ($m \geq 3$)	PROVED $m = 2$ separately, CONJ $m \geq 3$
vertex	multi	adjacency count, so the simple digraph value	convention
vertex	hyper	$(1 + o(1))(m-1)n^2/(4(r-1))$	LEADING TERM exact value open

PROVED holds for every n . **CONJ** conjectured formula, with a matching lower bound proved and the small cases exhausted.

LEADING TERM the n^2 term is a theorem, the exact value is not.

1 Hypergraph vertex problem: settled at $m = 2$ and sharp at $m = 3$ in its construction range

a universal $m = 3$ upper bound follows from a new incidence-graph rank bound, with equality proved for $r \geq 3$ when $2 \leq \binom{n-2}{r-2}$

2 One-way simple graphs: an unconditional quadratic bound

$|A| \leq \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-1)$, so the leading constant and the order of the next term are theorems, not conjectures

3 One-way multigraphs, solved for every n and every m

$L_m^{\text{dir}}(n) = (m-1) \max(2(n-1), \lfloor n^2/4 \rfloor)$, by deleting a sparse subgraph that alone remembers who can reach whom, once per unit of $m-1$

What is left

- ■ The undirected *vertex* problem at $m \geq 6$, open since 1974
- ■ The structural decomposition that would turn the directed conjecture into a theorem
- ■ The exact values behind the leading terms now settled, one-way and for hyperedges alike

The search proposes a lower bound. A proof or a finished exhaustion disposes the upper bound.

Only when the two meet is a value called settled.